Problems

In 2000, Rocco Baldelli was a first-round draft pick by the Tampa Bay Rays baseball team. Two years later, his early success in professional baseball was followed by disabling muscle injuries and fatigue. His mysterious illness was traced to a disorder called mitochondrial myopathy. This diagnosis brought not only personal clarification to one set of identical twin brothers, but also national awareness of an array of similar disorders that affect thousands of children and adults regardless of age, gender, race, or position in society. For 14 years the twin brothers competed on equal terms until Rocco began to suffer loss of visual acuity and muscularity. Diagnosis of a mitochondrial disorder explained the cause and forecasted supportive treatment, but also raised some important questions.

1. The parents and other relatives of the twins are apparently healthy, so how could this genetic disorder have arisen?
2. How could one identical twin have an inherited mitochondrial disorder and not the other?
3. Will the unaffected brother be guaranteed continued good?

1. While parents and other relatives are all healthy, but mother's eggs may accumulate the mitochondrial DNA mutations which happened only in a small number of cells in the mother. The mother's phenotype is not affected, whereas in the offspring, zygotes with mitochondrial mutations can develop with different proportions of mutated mtDNA distributed, causing him obvious disease phenotype.
2. Monozygotic twins are the result of a fertilized egg splitting early in development and eventually developing into two separate embryos. During the first several divisions, cytoplasm may be separated with one child have the majority of the mutated mtDNA and the other doesn't, resulting one sick and the other healthy.
3. No. The healthy one may have low number of mitochondria that do not affect the phenotype at present. As age increases, mutant mtDNA will accumulate, leading to the emergence of disease phenotypes.